

I. THE IMPORTANCE OF VISUALIZATION IN AI

A picture is worth a thousand rows of data. In the Machine Learning pipeline, visualization serves three main purposes:

1. **Exploratory Data Analysis (EDA):** Detecting outliers, trends, and correlations in raw data.
2. **Model Diagnostics:** Visualizing loss curves, accuracy, and confusion matrices to see if the model is learning.
3. **Communication:** Presenting complex findings to non-technical stakeholders (Business Intelligence).

II. MATPLOTLIB: THE FOUNDATIONAL LIBRARY

Matplotlib is the "grandfather" of Python visualization libraries. It provides low-level control over every element of a plot (axes, labels, colors, markers).

2.1. Basic Plot Types

- **Line Plots:** Best for showing trends over time (e.g., stock prices or model training loss).
- **Scatter Plots:** Ideal for showing the relationship (correlation) between two numerical variables.
- **Bar Charts:** Used for comparing quantities across different categories (e.g., number of students per major).
- **Histograms:** Essential for understanding the "Distribution" of a variable (e.g., the spread of exam scores).

2.2. The Subplot System

For complex reports, Matplotlib allows you to create a "Figure" containing multiple "Axes" (subplots). This is useful for comparing different features of a dataset side-by-side.

III. SEABORN: STATISTICAL DATA VISUALIZATION

Seaborn is built on top of Matplotlib but provides a high-level interface and much more attractive default styles. It is specifically designed to work with Pandas DataFrames.

3.1. Advanced Statistical Plots

- **Heatmaps:** Frequently used to visualize "Correlation Matrices." It helps engineers decide which features are redundant and should be removed.
- **Box Plots (Whisker Plots):** The best tool for **Outlier Detection**. It shows the median, quartiles, and any data points that fall outside the expected range.
- **Violin Plots:** Similar to box plots but also shows the probability density of the data at different values.
- **Pair Plots:** Automatically creates a grid of scatter plots for every numerical variable in a dataset—perfect for initial EDA.

IV. DATA STORYTELLING PRINCIPLES

To make your DTU Capstone Project professional, follow these visualization rules:

1. **Always Label Axes:** A plot without x and y labels is meaningless.
2. **Choose the Right Scale:** Use log scales for data with extreme ranges (e.g., population or wealth).
3. **Color with Purpose:** Use color to highlight the most important data point, not just for decoration.
4. **Avoid "Chart Junk":** Remove unnecessary gridlines or 3D effects that distract from the data.

V. PRACTICAL LAB: VISUALIZING THE DATASET

Python

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import pandas as pd
```

1. Setting the style

```
sns.set_theme(style="whitegrid")
```

2. Scatter plot with regression line (Seaborn)

```
plt.figure(figsize=(10, 6))
```

```
sns.regplot(x='study_hours', y='exam_score', data=df)
```

```
plt.title("Correlation: Study Hours vs Exam Scores")
```

```
plt.show()
```

3. Correlation Heatmap

```
plt.figure(figsize=(12, 8))
```

```
correlation_matrix = df.corr()
```

```
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
```

```
plt.title("Feature Correlation Matrix")
```

```
plt.show()
```